

VEHICLE SERVICE MANAGEMENT SYSTEM

by : B.Divya
email : mail2divya.bathini@gmail.com

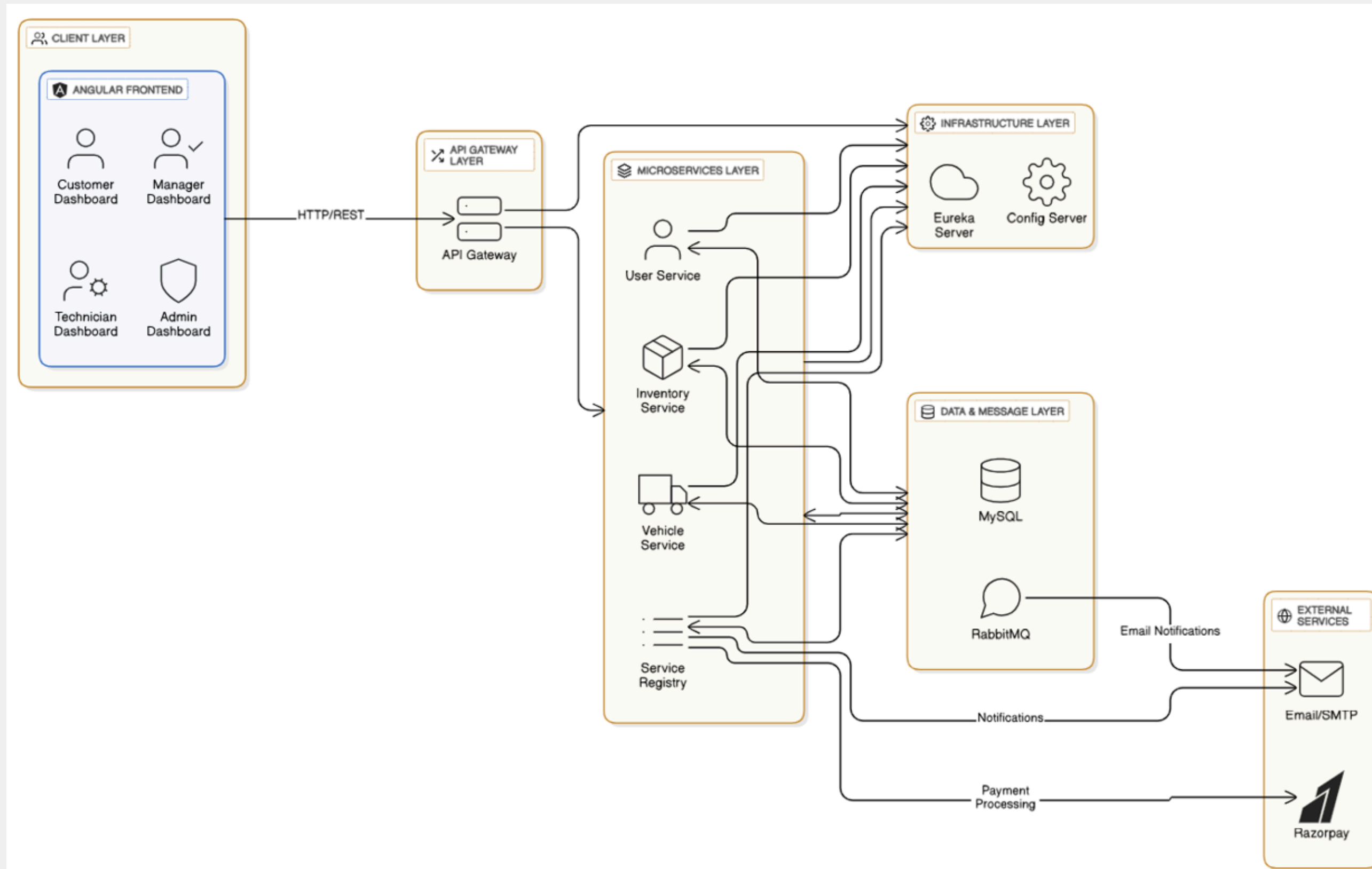
PROJECT OVERVIEW

Automobile service centers use manual or semi-automated systems, making it hard to track service history, schedule technicians, manage spare parts, and monitor workload and revenue. This lack of a centralized system leads to service delays, poor visibility, and reduced customer satisfaction.

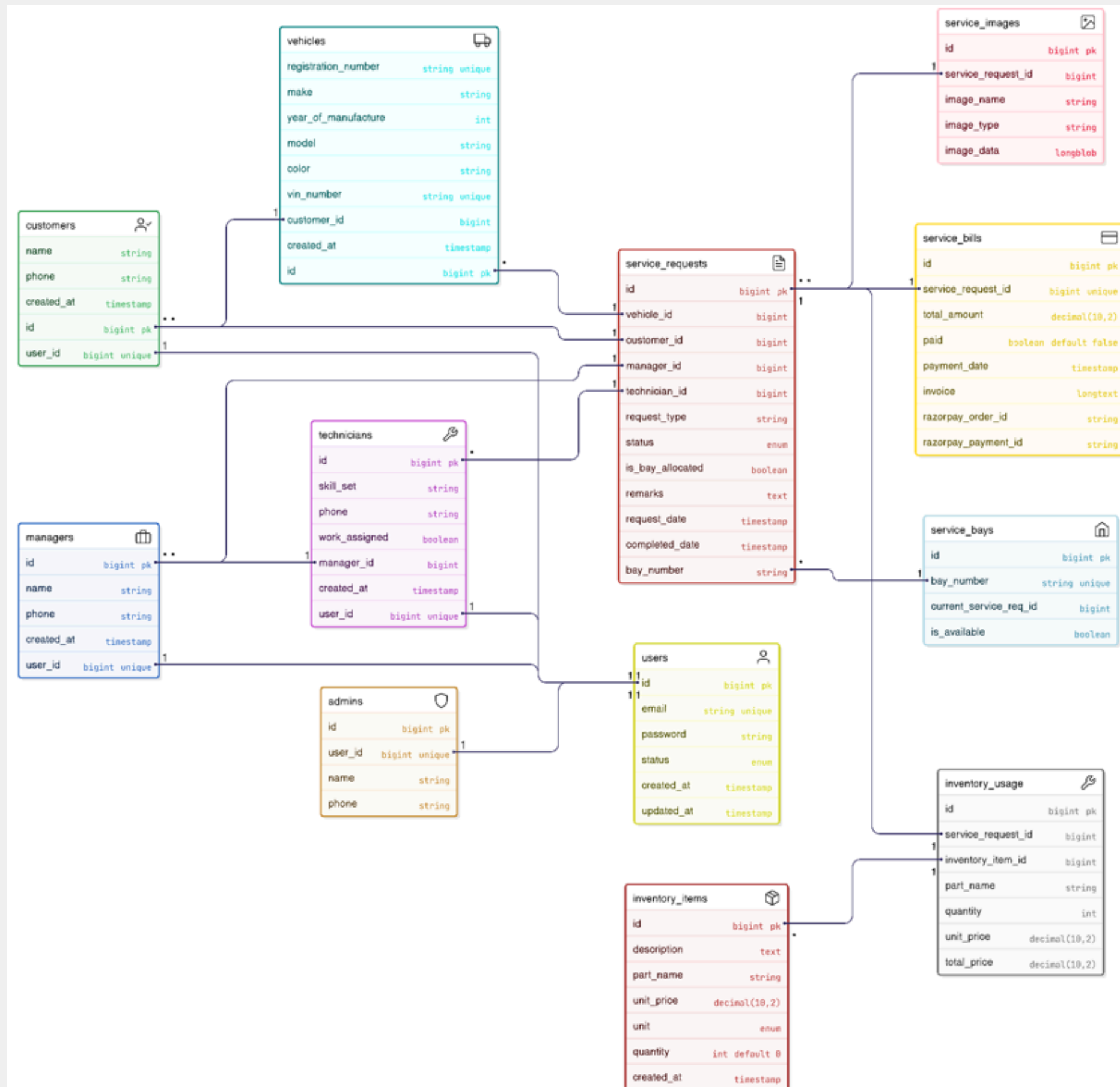
Solution :

- Centralized platform for managing vehicle services and history
- Automated technician and service bay assignment
- Real-time inventory and spare parts tracking
- End-to-end service lifecycle management
- Role-based dashboards for all users
- Automated billing, invoicing, and email notifications

MICROSERVICES ARCHITECTURE



DATABASE SCHEMA



USER ROLES & RESPONSIBILITIES

ADMIN	Allow managers to Login ,Add Bay Areas, Assign Customers to the manager
MANAGER	Allow Technicians to work under that manager ,Allocate technicians, Manage Inventory
TECHNICIAN	Do the task assigned by manager and admin
USER	Manage their view, Pay for that service

SERVICE REQUEST WORKFLOW

Lifecycle Flow:

PENDING → ASSIGNED → MANAGER_ASSIGNED → IN_PROGRESS → COMPLETED

Step-by-Step:

1. Customer create a service request with Vehicle Details.
2. Admin can assign the task to manager.
3. Manager adds labour cost and assign the task based on request to technician
4. Technician take the task and complete the work.
5. Bill will be autogenerated and customer can pay using razorpay.

KEY FEATURES IMPLEMENTED

Feature Module	Functionality	How It Works (Implementation)
Account registration	Approval/Rejection by Admin	Admins can approve or reject managers and technicians by verifying their identities.
Smart Assignment	Workload Balancing	System automatically blocks assignment if a Technician has reached their daily job capacity (e.g., 5 jobs/day).
Payments	Razorpay Integration	Secure payment gateway with Signature Verification (HMAC-SHA256) to ensure transaction authenticity before updating invoice status.
Analytics	Technician Statistics	Real-time dashboards showing Active Workload (Current Jobs) vs. Performance History (Completed Jobs) for Manager oversight.
Invoice	Download Bill	After the completion of service request by technician , automatic bill will be generated which can be downloaded by the customer.

DOCKER

<



vehicleservicemanagementsystem
C:\Users\BATHINI HEMASREE\Documents\workspace-...

View configurations







	mysql:8.0 3307:3306				
	notifica... vehicleservic				
	api-gate... vehicleservic 8080:8080				
	service-... vehicleservic				
	vehicle-... vehicleservic				
	user-ma... vehicleservic				
	config-s... vehicleservic 8888:8888				
	eureka-... vehicleservic 8761:8761				

at
org.springframework.amqp.rabbit.connection.A
bstractConnectionFactory.
createBareConnectio
n(AbstractConnectio
nFactory.java:590)
... 10 common
frames omitted
2026-01-07 01:32:18 -
Restarting
Consumer@38c78b33:
tags=[[]],
channel=null,
acknowledgeMode=AUTO
local queue size=0
2026-01-07 01:32:18 -
Closing Rabbit
Channel: null
2026-01-07 01:32:18 -
Attempting to connect
to: [rabbitmq:5672]
2026-01-07 01:32:24 -
Resolving eureka
endpoints via
configuration

CI/CD PIPELINE

 **Jenkins** / Vehicle Management Syst... / #28 / Console Output

```
Container config-server Healthy
Container notification-service Starting
Container service-registry Started
Container user-service Started
Container notification-service Started
Container vehicle-service Started
Container inventory-service Started
Container api-gateway Started
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Declarative: Post Actions)
[Pipeline] echo
Pipeline completed successfully!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // withCredentials
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

SUMMARY

- Microservices Architecture with 8 independent services
- Secure JWT-based Authentication & Role-Based Access Control .
- Complete Service Lifecycle Management (Pending → Assigned → Manager_Assigned -> In Progress → Closed).
- Smart Workload Balancing & Real-time Bay Conflict Resolution.
- Real-time Email Notifications via RabbitMQ
- Automated Billing & Dynamic Invoice Download Generation.
- Razorpay Payment Integration.
- CI/CD Pipeline with Jenkins, Docker & SonarCloud.

THANK YOU